

TEAM HYPERLINK
NANDISHWAR EN
DIVYA VS

AGRILINK-OR

**Deterministic Decision Support
for Farmer Producer Organisations**

Where to sell. | **When** to sell. | **To whom** to sell.

No machine learning — every number traceable.

REAL AGMARKNET DATA

NO ML

DETERMINISTIC

TESTED



THE PROBLEM

The core decision
that drives profitability
yet remains unresolved.

SELL TODAY OR HOLD?
IF HOLD, WHERE AND WHEN?

PRICE TRANSPARENCY

Ignores freight, cess, handling, and transit shrinkage

Highest quoted mandi price $\neq$ highest net cash in hand

WHAT THIS

Suboptimal mandi choice

Mistimed sales harming profitability

Fragmented smallholder lots too small for institutional buyers

THE 3 W'S COMMERCIAL METHODOLOGY

W	DECISION CARD	DECISION CARD	QUESTION	ENGINE
	W1 - WHERE	SELL AT	Which mandi leaves the most cash in hand after freight, cess, handling, and transit shrinkage?	M2 Net in hand Ranking (spatial arbitrage)
	W2 - WHEN	HOLD	Is holding profitable, on what date does it break even, and what's the optimal liquidation schedule?	M3 Break-even + CBC Milp
	W3 - TO WHOM	BULK ORDER	Which farmers' lots fill an institutional contract with 3% surplus and no single farm exceeding 40%?	M4 Bounded Knapsack DP cross-check

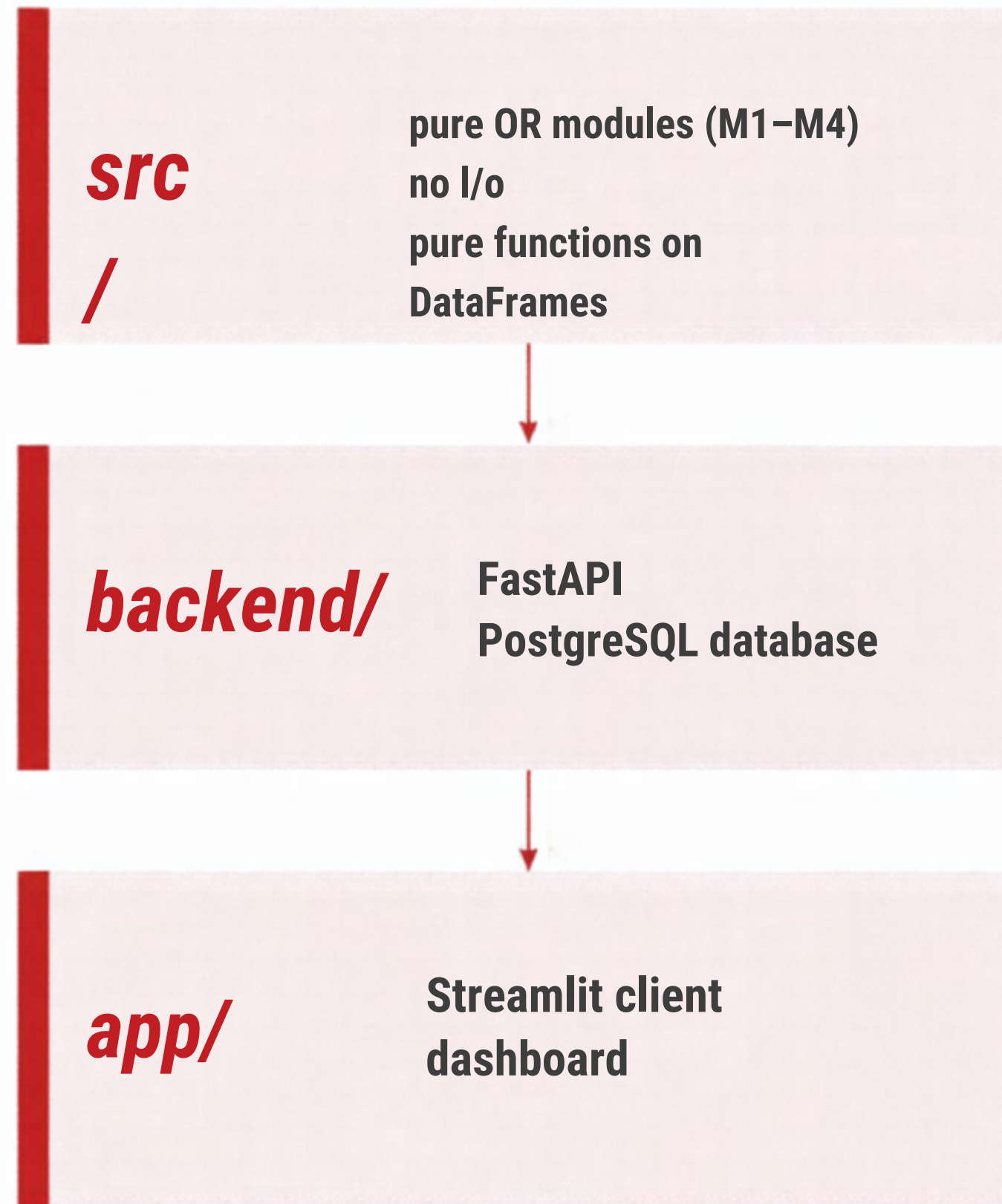
Where maximises immediate net cash, When times the seasonal peak against storage decay, To Whom packs smallholder lots into institutional supply.

SYSTEM ARCHITECTURE

Three layers.
Pure computation.
Deterministic outcomes.

**IDENTICAL MATH
REGARDLESS OF
DATA SOURCE**

Parquet files or
database rows –
same results.



API SUMMARY

11 operations across 10 API paths

/api/decision

-api/m1

-api/m2

-api/m3

-api/m4

/api/eligible-mandis

/api/params

/api/export

/api/health

/api/meta

validated inputs yield standard

400/422/503

error responses instead of crashing

DATA FOUNDATION

KARNATAKA BASELINE SLICE	DATA PROVENANCE			
	ROWS	MANDIS	DISTICTS	CROPS
	14,352	72	21	5
PAN-INDIA NATIONAL LAYER	Data sourced from Agmarknet dump			
	Cleaned and geocoded			
	No synthetic fallback – unsupported computations yield no guesswork			
KARNATAKA BASELINE SLICE	DATE RANGE			
	2023-06-06 to 2025-06-11			
PAN-INDIA NATIONAL LAYER	ROWS	STATE	GEODED MANDIS	
	727,050	\$ 26	1,611	
KARNATAKA BASELINE SLICE	DATE RANGE			
	2023-06-06 to 2025-06-11			

METHODOLOGY DETAIL

Four deterministic modules powering **AgriLink-OR**

M1 STL 52-week LOESS decomposition ±2' volatility bands GLUT/SPIKE z-score flags Seasonal price path $p(d)$	M2 NIHR Cost-walk ranking across mandis Karnataka / Intestate / Pan-India (26-state) scope modes Diesel-freight and road-circuitry modeling	M3 MILP Continuous value curve $V(t)$ Break-even day scanner CBC solver multi-tranche liquidation schedule	M4 Knapsack Bounded knapsack allocation Independent 0/1 subset-sum DP cross-check 40% farm-concentration cap
✓ 43 automated tests passing	zero ML	fully deterministic	fully auditable

RESULTS: REPRODUCIBLE ONION SCENARIO

(2025-06-11)

W1 • M2					W2 • M3					W3 • M4					SEASONAL PATTERN				
SELL AT					HOLD					BULK ORDER					Peak window: November (index 1.56, +56%)				
Location: Channarayapatna					157 days holding period Break-even on day 4					400 qtl from 8 farmers 0 qtl surplus					Trough window: May (index 0.62, -38%)				
Net •2,756/qtl (211 km)					Gain •2,177/qtl MILP objective: •932,893					DP cross-check agrees									
COST-WALK (•/QTL)																			
₹3,000	₹13	₹134	₹98	=	₹2,756														
Board	Transit loss	Freight	Cess & handling		Net														

Three answers, one scenario, fully reproducible from 6,330 in-view price rows across 43 mandis.